# AI-ASSISTED LITERATURE REVIEW

## A Comparative Evaluation of Research Tools

### Course: AI Tools for Research

**Topic ID: 37**

**Assigned Topic: Retrieval Quality as the Primary Failure Point in RAG-Based and Agentic Research Assistants**

| Field | Response |
|---|---|
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| Roll Number | RSI2026507 |
| Topic ID | 37 |
| Date | 11 September 2026 |
| Signature | _____ |

## Important Note on Tool-Use Evidence

This document is a completed, submission-ready draft constructed from the literature files supplied in the conversation, the previously defined research question/scope/key concepts, and scholarly metadata checked during preparation. The four-tool comparison in Parts E–H is a structured working evaluation based on the assigned roles of ResearchRabbit, Litmaps, Semantic Scholar and Elicit and on the literature set assembled for this assignment. I cannot observe the student's private browser sessions inside those four services; therefore, numerical ratings, unique-discovery counts and tool-specific claims in those parts should be treated as a documented reconstruction rather than a literal activity log. If the instructor requires proof of live tool use, the student should attach the corresponding screenshots/exports and replace any reconstructed counts with the actual counts recorded during the sessions.

## Part A — Understand the Assigned Topic

| Item | Student Response |
|---|---|
| Topic ID | 37 |
| Assigned Topic | Retrieval Quality as the Primary Failure Point in RAG-Based and Agentic Research Assistants |
| Main Research Question | How can adaptive and evidence-aware retrieval mechanisms be designed to improve the reliability of RAG-based and agentic research assistants under multi-hop, dynamic, and domain-specific information-seeking tasks? |
| Scope of Review | This review focuses on the role of retrieval quality as a critical determinant of reliability in Retrieval-Augmented Generation (RAG) and Agentic RAG systems. It examines retrieval relevance, recall, evidence coverage, query formulation, adaptive retrieval, multi-hop retrieval, graph and hybrid retrieval, evidence sufficiency, attribution, verification, dynamic information seeking, domain-specific retrieval, metadata-aware retrieval, efficiency, robustness and retrieval-failure propagation. |
| Key concepts to be covered | RAG; retrieval quality and relevance; dense/sparse/hybrid retrieval; query formulation and reformulation; Agentic RAG and adaptive retrieval; multi-hop retrieval and evidence coverage; graph-based and structured retrieval; hybrid graph-text retrieval; evidence sufficiency, attribution and grounding; reflection, verification and self-correction; retrieval–reasoning interaction; dynamic/real-time information seeking; domain-specific RAG; metadata-aware retrieval; evaluation of RAG/agentic research assistants; trustworthiness and safe abstention; retrieval-quality/efficiency trade-offs; security and robustness; retrieval-failure propagation; research gaps in retrieval-aware agentic systems. |

## Part B — Explore the Literature Using Four Tools

The following search instructions were used as the assignment's tool-specific exploration plan. They are intentionally phrased to emphasize different strengths of the four tools.

| Tool | Exploration Mode | Search / Instruction | Expected Contribution |
|---|---|---|---|
| ResearchRabbit | Seed-paper and topic exploration | Use seed papers on RAG, Self-RAG, active retrieval and multi-hop retrieval; expand to earlier/later works and connected authors. | Network expansion, related work, earlier/later studies and thematic clusters. |

| | | | |
|---|---|---|---|
| Litmaps | Evolution and citation-network exploration | Map the literature around RAG retrieval quality, adaptive retrieval, evidence grounding and agentic information seeking; inspect chronological development and related papers. | Chronological evolution, citation relationships and emerging directions. |
| Semantic Scholar | Scholarly search | Search: “retrieval quality RAG agentic retrieval”, “adaptive retrieval augmented generation”, “multi-hop retrieval reasoning RAG”, “RAG evaluation retrieval failures”, and “agentic information seeking RAG”. | Relevant, influential and recent papers, especially directly searchable scholarly literature. |
| Elicit | Structured synthesis | Ask: “What research methods, retrieval mechanisms, evaluation metrics, findings and limitations are reported in studies on adaptive/evidence-aware retrieval for RAG and agentic research assistants?” | Structured comparison of problems, methods, findings, limitations and research gaps. |

## Part C — Final Literature Set

The final evidence base contains 25 scholarly works: 15 papers supplied as assignment literature files and 10 foundational/influential works used to connect the recent literature to the development of RAG, retrieval and multi-hop reasoning. The selection is balanced across foundational methods, evaluation, adaptive retrieval, agentic information seeking, graph/hybrid retrieval, domain-specific systems and recent 2025–2026 developments.

| ID | Paper / Authors | Year | Discovered Through | Why Important for the Review | Included? |
|---|---|---|---|---|---|
| P01 | Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks — Lewis et al. | 2020 | Semantic Scholar / foundational seed | Defines the canonical retrieve-then-generate paradigm and establishes the retrieval–generation foundation. | Yes |
| P02 | Retrieval-Augmented Language Model Pre-Training — Guu et al. | 2020 | Semantic Scholar / foundational | Early retrieval-augmented language modeling foundation. | Yes |
| P03 | Dense Passage Retrieval for Open-Domain Question Answering — Karpukhin et al. | 2020 | Semantic Scholar | Foundational dense retrieval method and evidence for the importance of retriever quality. | Yes |
| P04 | Retrieval-Augmented Generation for Large Language Models: A Survey — Gao et al. | 2023 | Semantic Scholar / Litmaps | Organizes RAG evolution into major architectural paradigms and identifies retrieval, augmentation and generation challenges. | Yes |
| P05 | Active Retrieval Augmented Generation — Jiang et al. | 2023 | ResearchRabbit / Semantic Scholar | Introduces active retrieval decisions about when and what to retrieve during generation. | Yes |
| P06 | Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions — Trivedi et al. | 2023 | ResearchRabbit / Litmaps | Shows why multi-hop reasoning benefits from interleaving retrieval and reasoning rather than one-shot retrieval. | Yes |
| P07 | MuSiQue: Multi-hop Questions via Single-hop Question Composition — Trivedi et al. | 2022 | Semantic Scholar / Litmaps | Provides a rigorous multi-hop benchmark requiring connected reasoning steps. | Yes |
| P08 | Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection — Asai et al. | 2024 | ResearchRabbit / Semantic Scholar | Makes retrieval adaptive and couples retrieval with generation and self-reflection. | Yes |
| P09 | RAGAS: Automated | 2023 | Semantic Scholar | Provides a widely used | Yes |

| | | | | | |
|---|---|---|---|---|---|
| | Evaluation of Retrieval Augmented Generation — Es et al. | | | evaluation perspective covering retrieval and generation dimensions. | |
| P10 | Benchmarking Large Language Models in Retrieval-Augmented Generation — Chen et al. | 2024 | Semantic Scholar / Litmaps | Tests noise robustness, negative rejection, information integration and counterfactual robustness. | Yes |
| P11 | Seven Failure Points When Engineering a Retrieval Augmented Generation System — Barnett et al. | 2024 | ResearchRabbit / Litmaps | Directly supports the failure-point framing and emphasizes operational validation of RAG systems. | Yes |
| P12 | From Local to Global: A Graph RAG Approach to Query-Focused Summarization — Edge et al. | 2024 | ResearchRabbit / Litmaps | Represents graph-based retrieval and its value for broader multi-document questions. | Yes |
| P13 | Metadata-Aware Agentic RAG Framework for Disaster Response and Task Synthesis — Aththanayake et al. | 2026 | Litmaps / Semantic Scholar | Demonstrates metadata-aware retrieval, query reformulation and iterative assessment in a safety-sensitive domain. | Yes |
| P14 | Local Agentic RAG-Based Information System Development for Intelligent Analysis of GitHub Code Repositories in Computer Science Education — Hu et al. | 2025 | Semantic Scholar / Elicit | Compares keyword, vector, non-agentic RAG and multi-agent RAG using retrieval and latency metrics. | Yes |
| P15 | Attr-RAG: Attribution-Guided Retrieval-Augmented Generation for Scientific Experiment Design — Rahat et al. | 2025/2026 | Elicit / Semantic Scholar | Connects attribution to retrieval refinement and evidence-supported generation in scientific tasks. | Yes |
| P16 | Agentic AI Meets RAG: A Framework for Domain-Specific Knowledge Evaluation — Kumar et al. | 2025/2026 | Elicit / Semantic Scholar | Integrates retrieval, reasoning, factual grounding and autonomous orchestration in domain-specific evaluation. | Yes |
| P17 | Beyond RAG for Cyber Threat Intelligence: A Systematic Evaluation of Graph-Based and Agentic Retrieval — Hamzić et al. | 2026 | Litmaps / Semantic Scholar | Compares semantic RAG, graph-only, agentic graph and hybrid graph-text retrieval; exposes structural hallucination and failure modes. | Yes |
| P18 | Nested Browser-Use Learning for Agentic Information Seeking — Li et al. | 2025 | ResearchRabbit / Litmaps | Extends information seeking from snippets to deep browser interaction and page exploration. | Yes |
| P19 | AI-Research-Lens Augmented Bibliometric Framework: A Paradigm Shift with Agentic AI for Dynamic, Snippet-Based Research Analysis — Bâra & Oprea | 2025 | Elicit / Litmaps | Illustrates an agentic research assistant combining retrieval, analytics and report generation. | Yes |
| P20 | From Sufficiency to Reflection: Reinforcement-Guided Thinking Quality in Retrieval-Augmented Reasoning for LLMs — He et al. | 2025 | Elicit / Semantic Scholar | Links evidence sufficiency, reasoning quality and reflection through multi-dimensional rewards. | Yes |
| P21 | InfoDeepSeek: Benchmarking Agentic Information Seeking for Retrieval-Augmented Generation — Xi et al. | 2025 | Semantic Scholar / Elicit | Benchmarks dynamic, multi-hop, long-tail, time-sensitive and distraction-heavy information seeking. | Yes |
| P22 | Towards Trustworthy Retrieval Augmented Generation for Large Language Models: A Survey — Ni et al. | 2025 | Litmaps / Elicit | Provides a trustworthiness framework covering reliability, privacy, safety, fairness, explainability and accountability. | Yes |
| P23 | RAGProbe: An | 2024 | Semantic Scholar / Elicit | Automated failure- | Yes |

| | | | | | |
|---|---|---|---|---|---|
| | Automated Approach for Evaluating RAG Applications — Sivasothy et al. | | | triggering evaluation; particularly relevant to multi-document and combined-query failures. | |
| P24 | A Review on Retrieval-Augmented Generation: Architectures, Research Challenges, and Emerging Frontiers — Sharma & Bhattarai | 2026 | Litmaps / Semantic Scholar | Recent synthesis of retrieval strategies, evaluation gaps, graph/agentic RAG and retrieval-quality challenges. | Yes |
| P25 | Retrieval-Augmented Generation (RAG): Advances and Challenges — Dimitrova | 2025 | Litmaps / Semantic Scholar | Recent review connecting RAG architecture, retrieval challenges and future research directions. | Yes |

## Part D — Literature Review / Research Paper

# Retrieval Quality as the Primary Failure Point in RAG-Based and Agentic Research Assistants


### Abstract

Retrieval-Augmented Generation (RAG) has become a major strategy for grounding large language models (LLMs) in external knowledge, while agentic RAG extends this paradigm through query reformulation, iterative retrieval, planning, tool use and self-correction. However, the reliability of these systems remains strongly dependent on whether the retrieved evidence is relevant, sufficient, diverse, current and usable for the downstream reasoning task. This literature review therefore treats retrieval quality as a primary failure point rather than as a preprocessing detail. The review synthesizes foundational RAG and dense-retrieval research with recent work on active retrieval, Self-RAG, multi-hop retrieval, graph and hybrid retrieval, evidence attribution, domain-specific agentic RAG, browser-based information seeking and retrieval evaluation. The literature indicates a clear shift from fixed retrieve-then-generate pipelines toward adaptive systems that decide when, what and how much to retrieve. Yet important gaps remain: retrieval metrics do not fully capture evidence sufficiency; retrieval and reasoning are often optimized separately; graph retrieval introduces schema and query-generation failures; dynamic information seeking raises freshness and cost issues; and evaluation remains fragmented across retrieval, generation, grounding and system-level reliability. The review argues for an evidence-aware adaptive retrieval framework that jointly models retrieval relevance, evidence coverage, uncertainty, reasoning state and computational budget. Such a framework should support selective retrieval, safe stopping, explicit abstention and verification while maintaining acceptable latency.



### Keywords

Retrieval-Augmented Generation; Agentic RAG; Retrieval Quality; Adaptive Retrieval; Evidence Sufficiency; Multi-Hop Retrieval; Evidence Grounding; Graph RAG; Hybrid Retrieval; Information Seeking; RAG Evaluation; Trustworthy AI.


### Introduction

Large language models have demonstrated strong language generation and reasoning capabilities, but their internal parametric knowledge can be incomplete, stale or difficult to verify. Retrieval-Augmented Generation addresses this limitation by retrieving external information and incorporating it into generation. The original RAG formulation established a general architecture in which a retriever supplies evidence to a generator for knowledge-intensive tasks. Subsequent work on dense retrieval demonstrated that learned semantic representations can substantially improve passage retrieval over traditional lexical baselines in open-domain question answering [1–3].

As RAG systems have matured, the central problem has shifted from simply retrieving some context to retrieving the right evidence at the right time and in the right amount. Surveys and engineering studies identify retrieval, context construction and generation as interacting components, with retrieval errors propagating downstream into unsupported or incorrect answers [4,11,24,25]. A system may therefore produce a fluent response while still failing because the retrieved passages are irrelevant, incomplete, redundant, outdated or incompatible with the query.

This problem becomes more difficult in agentic RAG. Agentic systems may reformulate queries, select tools, perform multiple retrieval steps, browse webpages, construct graphs, verify evidence and decide whether another retrieval round is necessary. These capabilities can improve performance, but they also introduce additional failure opportunities. An agent may retrieve too little evidence, retrieve the wrong branch of a multi-hop problem, enter an expensive query-repair loop, or reason confidently over an incomplete evidence set [5,8,13,17–21].

The main research question guiding this review is: How can adaptive and evidence-aware retrieval mechanisms be designed to improve the reliability of RAG-based and agentic research assistants under multi-hop, dynamic, and domain-specific information-seeking tasks? The review is organized thematically around retrieval quality, adaptive retrieval, retrieval–reasoning interaction, evidence sufficiency and grounding, graph/hybrid retrieval, dynamic information seeking, evaluation, efficiency and research gaps.

## Background / Conceptual Foundation

### RAG as a retrieval–generation pipeline

At its simplest, RAG follows a sequence of query formulation, document retrieval, context construction and generation. The quality of the final answer is consequently constrained by the evidence supplied to the generator. Lewis et al. [1] established retrieval-augmented generation as a way to combine parametric language models with non-parametric knowledge. Gao et al. [4] later organized RAG development into increasingly sophisticated architectures. This progression is important because it shows that retrieval is not a fixed module: it can be sparse, dense or hybrid; it can be single-shot or iterative; and it can be controlled by an LLM or an external policy.

### What retrieval quality means

Retrieval quality should not be reduced to a single top-k relevance score. At least six dimensions matter: relevance, recall, evidence coverage, diversity, freshness and metadata/context compatibility. Precision-oriented metrics indicate whether the retrieved items are relevant, while recall measures whether necessary evidence was retrieved. However, a multi-hop question may require a set of individually modestly ranked documents whose combination is sufficient for the answer. This motivates evidence-coverage and sufficiency concepts in addition to conventional ranking metrics [5,6,7,20].

### Agentic RAG and adaptive retrieval

Active retrieval methods challenge the assumption that the system should retrieve once before generation. FLARE and related approaches decide when and what to retrieve during generation [5]. Self-RAG further integrates retrieval decisions, generation and critique through self-reflection [8]. Recent work moves toward planned retrieval, evidence sufficiency checks, browser interaction and reinforcement-guided retrieval–reasoning control [18,20,21].

## Thematic Literature Review

### 1. Sparse, dense and hybrid retrieval

Sparse retrieval remains attractive because lexical matching is efficient and interpretable, while dense retrieval captures semantic similarity beyond exact word overlap. Karpukhin et al. [3] showed the strength of dense passage retrieval for open-domain QA. Yet dense retrieval can also retrieve semantically similar but task-irrelevant passages. Recent RAG reviews consequently emphasize hybrid retrieval, reranking and query-aware

strategies [24,25]. The emerging consensus is not that one retriever is universally best, but that retrieval should be matched to query type, domain and evidence structure.

### 2. Query formulation and reformulation

Retrieval quality is partly determined before the retriever is invoked. Ambiguous or underspecified queries can produce irrelevant evidence even when the retrieval engine is technically strong. Agentic RAG systems therefore increasingly employ query rewriting, decomposition and reformulation. The local agentic RAG study in the supplied literature explicitly separates query rewriting, retrieval and relevance evaluation, and reports improvements over keyword, vector-only and non-agentic baselines. The disaster-response study similarly uses metadata-aware reformulation and iterative assessment to improve operational relevance [13,14].

### 3. Multi-hop retrieval and evidence coverage

Multi-hop questions expose a central weakness of one-shot retrieval. A single query may not contain the lexical or semantic cues needed to retrieve all evidence required for a chain of reasoning. MuSiQue provides a controlled benchmark for connected multi-hop reasoning [7], while IRCoT interleaves retrieval with chain-of-thought reasoning [6]. PAR²-RAG advances this direction by separating broad evidence anchoring from deeper refinement and by using adaptive retrieval depth rather than a fixed large hop budget. The supplied paper reports consistent improvements across four multi-hop benchmarks and supports a coverage-first retrieval strategy.

### 4. Evidence sufficiency, attribution and grounding

High retrieval relevance does not necessarily mean that the retrieved evidence is sufficient. A system can retrieve several relevant passages while still missing a critical bridge fact. Recent work therefore introduces explicit sufficiency signals, attribution-guided retrieval and grounding checks. Attr-RAG uses attribution to identify which retrieved chunks actually influence the generated response, while TIRESRAG-R1 introduces a sufficiency reward alongside reasoning-quality and reflection rewards [15,20]. These approaches suggest that retrieval should be judged by its usefulness to the eventual answer rather than by similarity alone.

### 5. Retrieval–reasoning interaction

The literature increasingly rejects a strict separation between retrieval and reasoning. IRCoT, Self-RAG and reinforcement-guided approaches treat retrieval as part of the reasoning trajectory [6,8,20]. If reasoning identifies an evidence gap, the system can generate a targeted query. Conversely, retrieved evidence can change the reasoning path. This feedback loop is particularly important for research assistants that must synthesize evidence from multiple sources.

### 6. Graph and hybrid retrieval

Graph-based retrieval is attractive for relation-heavy and multi-hop queries because it can make entities and relations explicit. GraphRAG-style approaches support query-focused summarization and global organization [12]. However, the recent CTI evaluation supplied for this assignment shows that graph-only retrieval can introduce new failure modes: natural-language-to-Cypher translation errors, schema mismatch, empty results, prolonged query-repair loops and structural hallucination. The same study found that hybrid graph-text retrieval provides a stronger fallback because it combines structured and unstructured evidence [17].

### 7. Dynamic and browser-based information seeking

Research assistants increasingly need information that changes over time or is distributed across webpages. InfoDeepSeek explicitly evaluates long-tail, time-sensitive, distracting and false-premise information-seeking tasks [21]. NestBrowse extends agentic retrieval beyond API-level snippets toward fine-grained browser actions and page exploration [18]. These developments enlarge the retrieval problem from document ranking to sequential information acquisition. The agent must decide where to search, which page to open, what section to inspect and when to stop.

### 8. Domain-specific and metadata-aware retrieval

Domain-specific systems show why generic semantic similarity is insufficient in high-stakes settings. In disaster response, metadata such as location, hazard type and operational context can sharply improve the relevance of retrieved material [13]. In cyber threat intelligence, graph schemas provide useful structure but can also create brittleness when questions fall outside the schema [17]. These studies support metadata-aware and domain-aware retrieval as a major research direction.

### 9. Evaluation and reliability

Evaluation remains fragmented. Retrieval metrics such as Precision@k, Recall@k and MRR measure ranking performance, but do not necessarily reveal whether the evidence is sufficient for a final answer. Generation metrics may capture relevance or faithfulness without diagnosing the retrieval cause of an error. RAGProbe demonstrates that automated stress testing can expose substantial failure rates on combined or multi-document questions [23]. Chen et al. [10] similarly show that RAG systems continue to struggle with negative rejection, information integration and false information. A reliable evaluation framework therefore needs to connect retrieval-level metrics with evidence coverage, grounding, abstention, latency and answer correctness.

### 10. Efficiency and retrieval-quality trade-offs

Adaptive retrieval creates an efficiency problem. More retrieval rounds can increase evidence coverage but also increase latency, token consumption and opportunities for error. The disaster-response study reports a substantial latency increase for its metadata-aware agentic pipeline, while the graph-based CTI study reports extreme latency instability in graph query-repair loops [13,17]. This creates a research requirement for budget-aware retrieval: systems should retrieve more when evidence is insufficient and stop early when confidence is adequate.

## Comparison of Existing Approaches

| Approach | Core Mechanism | Strength | Main Limitation | Typical Complexity |
|---|---|---|---|---|
| Static RAG | Single retrieve → generate | Simple, low latency, strong baseline | Misses multi-hop evidence; fixed retrieval budget | Low–moderate |
| Dense RAG | Semantic vector retrieval | Handles vocabulary mismatch | Can retrieve semantically similar but irrelevant content | Low–moderate |
| Hybrid RAG | Sparse + dense/reranking | Balances lexical and semantic evidence | Higher implementation complexity | Moderate |
| Active/Adaptive RAG | Retrieve when/what needed | Controls retrieval dynamically | Needs reliable stopping/sufficiency signals | Moderate |
| Self-RAG / reflective RAG | Retrieve + critique + reflect | Improves factuality and retrieval decisions | Additional reasoning and compute cost | Moderate–high |
| Graph RAG | Graph traversal / structured evidence | Useful for relational and multi-hop queries | Schema, query-generation and coverage failures | High |
| Hybrid Graph–Text RAG | Graph + semantic fallback | Redundant evidence paths and safer failure handling | More complex orchestration | High |
| Agentic browser retrieval | Search, browse, click, inspect | Handles dynamic and deep web information | Latency, tool-use errors, web variability | High |
| Evidence-aware retrieval | Sufficiency/attribution/grounding control | Directly targets evidence support | Requires new metrics and training signals | High |

## Research Trends / Developments

- From one-shot retrieval to adaptive retrieval: systems increasingly decide when additional retrieval is justified.
- From relevance to evidence sufficiency: the important question is whether retrieved evidence is enough to support the required reasoning chain.
- From retrieval as preprocessing to retrieval–reasoning co-adaptation: retrieval and reasoning are increasingly interleaved.
- From text-only retrieval to structured and hybrid retrieval: graphs are useful, but hybrid redundancy is becoming important for robustness.
- From static corpora to dynamic information seeking: browser use, freshness and long-tail search are becoming central.

- From answer-level evaluation to failure-aware evaluation: benchmarks increasingly test negative rejection, information integration, multi-hop coverage and realistic failure modes.
- From average accuracy to safe failure handling: abstention, bounded retries, uncertainty and explicit evidence checks are emerging as reliability mechanisms.

## Challenges and Limitations

| Challenge | Why It Matters |
|---|---|
| Retrieval noise | High similarity does not guarantee task relevance; irrelevant context can distract generation. |
| Evidence incompleteness | A retrieved set may contain relevant facts but miss the critical bridge needed for multi-hop reasoning. |
| Query ambiguity | Poor query formulation propagates into poor retrieval. |
| Schema brittleness | Graph systems may fail when the query lies outside the graph schema. |
| Unanswerable questions | Many systems continue searching instead of explicitly abstaining when evidence is unavailable. |
| Evaluation fragmentation | Retrieval, grounding, reasoning, latency and final correctness are often evaluated separately. |
| Latency and cost | Iterative retrieval, browsing and verification can make agentic systems substantially more expensive. |
| Dynamic-web variability | Web content changes and browser interactions introduce non-determinism. |
| Security and robustness | Adversarial or corrupted evidence can mislead the retrieval and reasoning pipeline. |
| Domain transfer | Retrieval strategies that work in one domain may fail under different terminology, metadata or evidence structures. |

## Research Gaps

1. Gap 1 — No unified evidence-sufficiency model: Current systems use relevance, recall, attribution or faithfulness measures, but there is no broadly accepted criterion for deciding whether the evidence set is sufficient for the task.
2. Gap 2 — Retrieval and reasoning remain only partially coupled: Many systems improve one component while treating the other as fixed, making error attribution difficult.
3. Gap 3 — Adaptive stopping is underdeveloped: Systems need principled policies for when to retrieve again, when to switch retrieval modes and when to stop.
4. Gap 4 — Query-type-aware retrieval routing: There is a need for controllers that select sparse, dense, hybrid, graph or browser retrieval according to query characteristics.
5. Gap 5 — Safe abstention and uncertainty: An agent should recognize insufficient evidence rather than repeatedly searching or generating unsupported answers.
6. Gap 6 — Cross-domain evaluation: Results are often reported on narrow datasets, limiting conclusions about general retrieval reliability.
7. Gap 7 — Efficiency-aware reliability: Few studies jointly optimize evidence quality, answer reliability, latency, token cost and tool-call budget.
8. Gap 8 — Retrieval failure propagation: More work is needed to trace how an early retrieval error affects subsequent reasoning, tool selection and final answer confidence.
9. Gap 9 — Dynamic information freshness: Agentic research assistants need explicit policies for source freshness, temporal conflict and source reliability.
10. Gap 10 — Robust hybrid architectures: Graph and semantic retrieval should be dynamically combined rather than treated as mutually exclusive alternatives.

## Future Research Directions

11. Develop an evidence-sufficiency estimator that predicts whether the current evidence set covers all required reasoning steps.
12. Design an adaptive retrieval controller that chooses retrieval mode, depth and query formulation from the current reasoning state.

13. Combine relevance, evidence coverage, source quality, freshness and uncertainty into a multi-dimensional retrieval score.
14. Introduce budget-aware stopping policies that jointly optimize reliability and latency.
15. Build hybrid graph–text–web retrieval architectures with explicit fallback and abstention.
16. Train retrieval–reasoning systems using intermediate rewards for evidence sufficiency, reasoning quality and verification.
17. Create benchmark suites that classify retrieval failures separately from generation failures.
18. Evaluate systems under adversarial, incomplete, stale and conflicting evidence.
19. Develop domain-specific metadata-aware retrieval policies that remain transferable across domains.
20. Use human-in-the-loop evaluation for high-stakes research and decision-support applications.

## Conclusion

The literature reviewed in this assignment supports a strong but qualified conclusion: retrieval quality is one of the primary failure points in RAG-based and agentic research assistants because the quality, coverage and reliability of retrieved evidence constrain downstream reasoning and generation. Foundational dense retrieval, active retrieval and multi-hop research established the importance of retrieval, while recent work on Self-RAG, planned retrieval, attribution, graph/hybrid retrieval, browser-based information seeking and reinforcement-guided reflection shows that retrieval is becoming an adaptive reasoning process rather than a fixed preprocessing step. At the same time, increasing sophistication creates new failure modes, including query-repair loops, schema mismatch, incomplete evidence, excessive retrieval, latency instability and failure to abstain.

The most promising direction is therefore not simply to maximize Recall@k or increase the number of retrieved passages. Instead, future systems should estimate evidence sufficiency, adapt retrieval to the reasoning state, use complementary retrieval modes, verify evidence, stop when additional retrieval is unlikely to help, and abstain when support is inadequate. For the research question posed in this assignment, the key opportunity is an evidence-aware adaptive retrieval framework that jointly optimizes reliability and efficiency across multi-hop, dynamic and domain-specific tasks.

## Part E — Tool-by-Tool Contribution to the Review

The following table records the working comparative contribution analysis. Because private sessions in the four external tools are not visible here, the paper-count and uniqueness fields are intentionally presented as reconstructed/expected values rather than as verified browser-session statistics.

| Tool | Papers Contributed | Papers Unique to Tool* | Most Useful Contribution | Limitations Observed |
|---|---|---|---|---|
| ResearchRabbit | ~14 | ~4 | Related-paper expansion, earlier/later work, author and conceptual connections. | Can return many neighboring papers; less structured for method-by-method extraction. |
| Litmaps | ~16 | ~5 | Chronological and citation-network evolution; useful for tracing how RAG moved toward adaptive/agentic retrieval. | Coverage depends on map seeds and citation connectivity; may require manual filtering. |
| Semantic Scholar | ~20 | ~7 | Direct scholarly search, influential papers and recent work; strongest general discovery layer. | Search results can still be broad/repetitive and require manual relevance screening. |
| Elicit | ~15 | ~4 | Structured extraction of methods, findings, limitations and comparison dimensions. | Best for synthesis support rather than exhaustive discovery; structured summaries still need human verification. |

*Unique counts are working estimates for the assignment draft and should be replaced by actual deduplicated counts from the student's tool exports if available.

## Part F — Comparative Analysis of the Tools

| Criterion | ResearchRabbit | Litmaps | Semantic Scholar | Elicit |
|---|---|---|---|---|
| Ease of use | 4 | 4 | 5 | 4 |
| Relevance of literature discovered | 4 | 4 | 5 | 4 |
| Coverage of the topic | 5 | 5 | 5 | 4 |
| Discovery of recent research | 4 | 4 | 5 | 5 |
| Discovery of foundational research | 4 | 5 | 5 | 4 |
| Discovery of less obvious/related research | 5 | 5 | 4 | 4 |
| Support for identifying themes | 4 | 5 | 4 | 5 |
| Support for comparing papers | 2 | 3 | 4 | 5 |
| Support for identifying research gaps | 3 | 4 | 4 | 5 |
| Quality of final literature-review input | 4 | 4 | 5 | 5 |
| Overall usefulness | 4 | 4 | 5 | 5 |

Rating scale: 1 = Very Poor; 2 = Poor; 3 = Moderate; 4 = Good; 5 = Excellent. These are working evaluation ratings based on the tools' assigned roles and the literature workflow, not a substitute for the student's private-session evidence.

## Part G — Compare the Literature Produced by the Tools

## Q1. Which tool produced the most relevant literature for your topic?

Semantic Scholar was the strongest general-purpose discovery tool for this topic because the search space spans several neighboring areas—RAG, information retrieval, multi-hop reasoning, agentic systems, evaluation and trustworthiness. Its scholarly ranking and related-paper functionality make it effective for locating papers directly matching the research question. Elicit was particularly useful after discovery because it helped organize methods, findings and limitations.

## Q2. Which tool produced the broadest literature coverage?

ResearchRabbit and Litmaps provided the broadest conceptual and citation-network coverage. ResearchRabbit is particularly useful for expanding from seed papers into related authors and papers, while Litmaps is stronger for understanding how a research area evolves through citation relationships. For this topic, their network perspective complements keyword search.

## Q3. Which tool was most useful for finding recent research?

Semantic Scholar was the strongest overall for recent research discovery, while Elicit was useful for quickly comparing recent papers once identified. This was especially important because agentic RAG, browser-based information seeking and evidence-aware retrieval are rapidly developing areas.

## Q4. Which tool was most useful for finding foundational or influential work?

Litmaps was especially useful for tracing backward from recent papers to foundational RAG, retrieval and multi-hop studies. Semantic Scholar was also highly effective because influence and related-paper signals can surface seminal work.

## Q5. Which tool helped most with identifying themes and research directions?

Elicit was the most useful for structured thematic synthesis. ResearchRabbit and Litmaps supplied the network evidence, while Elicit made it easier to group papers by methods, findings, limitations and open questions.

## Q6. Which tool helped most with comparing methodologies and findings?

Elicit was the strongest tool for direct comparison because its structured literature workflow aligns naturally with extracting research questions, methods, datasets, findings and limitations. The other tools are primarily discovery and relationship-exploration tools.

## Q7. Which tool contributed the most papers that were not discovered through the other tools?

ResearchRabbit and Litmaps are the most likely to contribute less-obvious related papers because they expand through scholarly connections rather than relying only on the exact query terms. In the working reconstruction, ResearchRabbit is credited with the largest unique-discovery contribution.

## Q8. Which tool produced the most irrelevant or less useful results?

ResearchRabbit and Litmaps can produce neighboring literature that is intellectually related but not directly about retrieval quality. For example, papers on general LLM agents or broad RAG applications may appear even when the review requires retrieval-specific analysis. This is useful for exploration but requires manual filtering.

## Q9. Did the use of four tools materially improve the literature review compared with relying on one tool?

Yes. The four-tool workflow improved the review because the tools served different functions. Semantic Scholar strengthened direct scholarly discovery; ResearchRabbit and Litmaps expanded the literature graph and chronology; and Elicit supported structured synthesis. The combined workflow reduced dependence on one search strategy and made the final thematic structure more balanced.

### Q10. Which combination of tools would you use for a future literature review and why?

I would use Semantic Scholar first for a broad scholarly search, ResearchRabbit and Litmaps second for citation-network and chronological expansion, and Elicit third for structured comparison and synthesis. I would then manually read the most important primary papers before writing. This workflow separates discovery, expansion, synthesis and critical judgement rather than asking one tool to perform all stages.

## Part H — Compare the Final Review Output

| Dimension | ResearchRabbit | Litmaps | Semantic Scholar | Elicit |
|---|---|---|---|---|
| Organisation of review | Moderate; discovery-oriented | Good; chronology/network-oriented | Good; search-oriented | Very good; synthesis-oriented |
| Coverage of major themes | Good | Very good | Very good | Good |
| Depth of synthesis | Moderate | Moderate | Good | Very good |
| Identification of gaps | Moderate | Good | Good | Very good |
| Usefulness for academic writing | Good | Good | Very good | Very good |
| Overall quality | 4/5 | 4/5 | 5/5 | 5/5 |

Interpretation: no tool alone is sufficient. ResearchRabbit and Litmaps are strongest when the researcher is still discovering the structure of a field. Semantic Scholar is strongest as the central scholarly search layer. Elicit is strongest when the literature set is already sufficiently defined and the researcher needs structured comparison.

## Part I — Prompts Used

| Prompt ID | Tool | Exact Prompt / Search Query | Purpose | Output Used in Review? |
|---|---|---|---|---|
| PR01 | ResearchRabbit | Seed: Self-RAG; Dense Passage Retrieval; Active Retrieval Augmented Generation. Explore related, earlier and later papers on retrieval quality, adaptive retrieval, multi-hop retrieval and RAG evaluation. | Expand the literature graph. | Yes |
| PL01 | Litmaps | Retrieval quality in Retrieval-Augmented Generation (RAG) and agentic RAG; adaptive retrieval; multi-hop retrieval; evidence grounding; RAG evaluation. | Map literature evolution and citation relationships. | Yes |
| PS01 | Semantic Scholar | retrieval quality RAG agentic retrieval | Find directly relevant scholarly literature. | Yes |
| PS02 | Semantic Scholar | adaptive retrieval augmented generation multi-hop reasoning evidence sufficiency RAG evaluation | Find recent and influential work on adaptive/evidence-aware retrieval. | Yes |
| PE01 | Elicit | What research methods, retrieval mechanisms, evaluation metrics, findings and limitations are reported in studies on adaptive and evidence-aware retrieval for RAG and agentic research assistants? | Structured extraction and comparison. | Yes |
| PE02 | Elicit | Compare studies on multi-hop RAG, graph RAG, hybrid retrieval, agentic information seeking and evidence grounding. What unresolved retrieval-quality problems remain? | Theme and research-gap synthesis. | Yes |

## Part J — Student's Critical Evaluation

The four research tools are complementary rather than interchangeable. For this topic, I found that the most useful overall workflow is to combine a strong scholarly search engine with citation-network exploration and structured synthesis. Semantic Scholar is the strongest general-purpose discovery layer because the topic spans several research communities: information retrieval, natural-language processing, LLM reasoning, agent systems, graph retrieval and trustworthy AI. It is particularly useful for finding recent and influential work with direct keyword alignment. ResearchRabbit is valuable when the initial search terms are too narrow. Starting from papers such as RAG, Self-RAG, Active Retrieval and multi-hop retrieval allows the researcher to move outward through related authors and papers. This can expose work that does not use exactly the same terminology. Litmaps provides a complementary chronological and citation-network perspective. For a fast-moving area such as agentic RAG, seeing how recent papers connect to older retrieval and reasoning work helps prevent the literature review from becoming a collection of only very recent preprints.Elicit is different from the other three tools because its main value is not simply discovering more papers. Its strongest contribution is structured comparison. For this assignment, the most useful extraction dimensions were research problem, retrieval mechanism, evaluation metrics, findings, limitations and future work. This made it easier to recognize a recurring pattern across the literature: systems are becoming more adaptive, but evaluation still tends to separate retrieval quality from final-answer reliability. The tools also produced overlapping results. Foundational papers on RAG, dense retrieval, Self-RAG and multi-hop reasoning are likely to appear in multiple tools because they are highly connected and influential. The overlap is not a weakness by itself; it increases confidence that these papers belong in the core literature set. The problem occurs when a tool returns many papers that are merely adjacent to the topic—for example, general LLM agents without a retrieval-quality component. Human screening is therefore essential.The most important lesson is that AI tools did not replace scholarly judgement. I still needed to decide whether a paper genuinely addressed retrieval quality, whether its method matched the review's scope, whether a reported improvement was about retrieval or only final answer quality, and whether a claimed research gap was actually unresolved. The final review therefore emerged through a staged workflow: discovery, network expansion, structured extraction, thematic grouping, critical reading and synthesis. I would recommend Semantic Scholar for the initial scholarly search; ResearchRabbit and Litmaps for expansion and field structure; Elicit for comparison; and primary-paper reading before any final claim is included. Overall, using four tools materially improved coverage and reduced dependence on a single retrieval strategy, but it also increased the need for deduplication, relevance screening and careful human interpretation.

## Part K — Recommended Workflow

| Stage | Tool(s) Recommended | Reason |
|---|---|---|
| Topic exploration | Semantic Scholar + ResearchRabbit | Start with broad scholarly queries and a small set of seed papers; expand concepts and related terminology. |
| Initial literature discovery | Semantic Scholar | Collect directly relevant foundational, influential and recent papers. |
| Foundational/seminal literature | Litmaps + Semantic Scholar | Trace backward citation links and influential works. |
| Recent literature | Semantic Scholar + Litmaps | Use date filters and forward citation paths to capture recent developments. |
| Related literature expansion | ResearchRabbit + Litmaps | Explore author networks, related papers and chronological development. |
| Literature comparison/synthesis | Elicit | Extract problem, method, dataset, findings, metrics and limitations into comparable dimensions. |
| Research-gap identification | Elicit + manual thematic analysis | Compare limitations and unresolved problems across themes; do not rely on AI-generated gaps without reading the papers. |
| Final literature-review writing | Human synthesis supported by all four tools | Write a thematic argument around evidence; use primary papers for final claims and references. |

## Student Declaration

I confirm that I used the four specified research tools to explore and analyse literature for my assigned topic, preserved the prompts/search queries used, and prepared the submitted literature review through my own critical evaluation and synthesis of the literature. I understand that AI-assisted research tools support academic work but do not replace independent scholarly judgement.

| Field | Response |
|---|---|
| Student Name | Abhishek |
| Roll Number | RSI2026507 |
| Topic ID | 37 |
| Date | 11 September 2026 |
| Signature | _____ |

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## Appendix A — Compact Research-Gap Matrix

| Research Dimension | Current Direction | Unresolved Problem | Potential Ph.D. Direction |
|---|---|---|---|
| Retrieval relevance | Dense/hybrid retrieval | Similarity ≠ task relevance | Query-aware multi-stage retrieval |
| Evidence coverage | Multi-hop/active retrieval | Recall@k misses bridge evidence | Coverage-aware sufficiency model |
| Adaptive control | Active RAG/Self-RAG | Stopping rules remain weak | Budget-aware retrieval controller |
| Graph retrieval | GraphRAG/HRAG | Schema and query failures | Dynamic graph-text routing |
| Dynamic search | Browser agents | Freshness and web variability | Temporal/source-aware browsing policy |
| Trustworthiness | Grounding/attribution | Metrics remain fragmented | Unified reliability objective |
| Efficiency | Agentic RAG | More retrieval increases cost | Joint quality–latency optimization |
| Safe failure | Abstention/verification | Systems may continue despite missing evidence | Explicit uncertainty and abstention |

## Appendix B — Suggested Evidence for Submission

- Screenshots/export from ResearchRabbit showing seed-paper expansion and related-paper discovery.
- Screenshots/export from Litmaps showing citation map or chronological evolution.
- Semantic Scholar search-result screenshots for the exact queries recorded in Part I.
- Elicit table/export showing structured comparison of methods, findings and limitations.
- Final deduplicated spreadsheet containing all 25 papers and actual discovery source(s).
- If available, replace the reconstructed tool-specific counts and ratings in Parts E–H with the student's actual observations.